Team gtx9090 · IBM Z Datathon 2026

Guardian Pay

Stopping “digital arrest” scams in the seconds before the money leaves

Theme: Real-Time AI for Critical Decisions · Built for IBM LinuxONE with IBM Granite

The victim is the real customer

A call that lasts hours

Someone posing as police, CBI or customs says a parcel in your name held drugs and keeps you on a video call.

Your own phone, OTP and PIN

The victim pays from their own device. Every check that asks “is this the real customer?” says yes.

Gone in minutes

The payment clears in milliseconds and is split across mule accounts long before anyone reports it.

Victims are often retired people, and the loss is often their life savings.

Fraud checks ask: is this the real customer?

We ask: is the real customer paying under pressure?

And we answer it inside the transaction, before the money moves.

Every payment, scored in under a millisecond

Signals

Leaving the app mid-payment, a pasted account number, freezing on confirm, an FD broken today, a young receiving account paid by many seniors

Hybrid engine

Instant rules, a coercion model on 18 signals, a PaySim pattern model, and the AI check of any message that came with the account

Decision

Under 40: paid

40–69: warning with reasons

70+: Guardian Pause

Median decision time 14 µs, p99 1.4 ms, measured scoring 20,000 payments one by one

Guardian Pause: two minutes that break the script

Hold, not refuse

The payment waits two minutes. A real emergency still goes through.

Saathi listens

An IBM Granite voice assistant asks, in Hindi, who asked for this payment, and warns plainly if it hears “officer” or “video call”.

Family confirms

The daughter’s phone shows the reasons and what Papa told Saathi. She taps Stop or lets it go.

“असली पुलिस या सीबीआई कभी फ़ोन पर पैसे नहीं मांगती।”

One customer’s answer protects everyone

10:02

Suresh, 66, Lucknow

Tells Saathi an officer is on a video call. He cancels.

10:02, same second

Account flagged bank-wide

R K Enterprises, a/c xx2906

10:05

Meena, 71, Pune

Calm, no stress signs. Paused anyway: the flag caught it.

Today this knowledge reaches the bank only when a 1930 complaint is filed, often hours later. Here the victim’s answer becomes a live signal.

Three trained models and IBM Granite

Model	Trained on	Job
Coercion model	1M generated payments, 8 personas incl. hard cases	Is the customer under pressure?
Transaction patterns	PaySim, 6.36M public mobile-money transactions	Does it look like known fraud?
Scam text model	12,156 real messages, 4 public datasets, 14 Indian languages	Scam type, urgency, blackmail
IBM Granite	watsonx.ai, and on LinuxONE via llama.cpp + zDNN	Saathi, message checker, case notes

Datasets: PaySim (Lopez-Rojas et al.), UCI SMS Spam, Indian cyber-scam call transcripts (Hinglish), scam/ham SMS in 14 Indian languages

Results on data the models never saw

98.5%

of digital-arrest payments paused

0.09% of genuine payments paused, 0.6% of real family emergencies

0.9996

PaySim AUC, trained on earlier days, tested on later ones

98.7% precision at score 80+

0.993

text model F1 on held-out messages

0.70–0.99 AUC on a dataset it never trained on

Coercion model: 800k train / 200k test, AUC 0.970. We report the cross-dataset numbers too, because they are the honest ones.

Why it belongs on IBM Z and LinuxONE

The payment is already there

Banks run payments on IBM Z, so the score is computed in the transaction, with no hop to another system.

Both sides of the payment

Sender behaviour and receiver activity live on one platform, which is what makes the community flag possible.

Granite on the Telum chip

Granite runs inside LinuxONE through llama.cpp with zDNN. No customer words leave the machine.

Trained where it runs

Our notebook trains all three models on LinuxONE in IBM’s AI image; production would serve them with Machine Learning for z/OS.

AI informs, people decide

Three layers read every sentence

Granite for meaning, the trained text model for learned patterns, and a negation-aware keyword net in English, Hinglish and Devanagari. Granite can raise the risk, never lower it below the other two.

No AI moves money

Saathi can recommend cancelling. The customer or the family contact makes the call.

It keeps working

watsonx down: Granite on LinuxONE. Both down: a scripted check, the text model and rules. Every Granite answer must be valid JSON or it is retried once, then replaced.

Honest limits

No public data of coerced payments exists, so that model learns from our generator. A bank would retrain it on its own cases.

Guardian Pay · team gtx9090

We don’t stop people paying. We give them two minutes and someone to ask.

Try it: pay from the phone app, talk to Saathi, and watch the console flag the mule account.

Code: github.com/prnvmishra/guardian-pay · Demo: youtu.be/BxpoHiEtRCo